Sara Ghasvarianjahromi, PhD

Machine Learning Research Engineer | Robust & Decentralized ML | Information Retrieval & RAG | LLMs

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Summary

Machine Learning Research E ngineer with a Ph.D. in Electrical Engineering and experience developing robust decentralized learning, semantic retrieval, and AI/ML-driven network optimization methods. First or corresponding author of research spanning information retrieval under token erasures, Byzantine-resilient distributed computation, and energy-aware 6G systems. Combines mathematical modeling with hands-on implementation in Python, PyTorch, Hugging Face, and Gurobi.

Technical Skills

Machine Learning: Semantic Retrieval, Retrieval-Augmented Generation (RAG), Text Embeddings, NLP, LoRA/PEFT, Deep Learning, Decentralized Learning, Adversarial Robustness

Programming & Libraries: Python, MATLAB, PyTorch, Hugging Face Transformers, scikit-learn, pandas, NumPy

Optimization & Domains: Gurobi, Convex Optimization, Probabilistic Modeling, Network Analytics, Resource Allocation

Tools: Git, Linux, VS Code, LaTeX

Experience

New Jersey Institute of Technology — Newark, NJ
Doctoral Research Assistant — *Jan 2022 – May 2026*

- Improved semantic-search reliability (ITW 2025, arXiv) under token erasures, measured by top-k retrieval accuracy, by developing a RAG-oriented pipeline with pretrained embeddings, cosine-similarity ranking, and token-level importance scoring.
- Evaluated importance-aware token redundancy on Google Natural Questions through Monte Carlo experiments using 29 Llama 3-generated query variations per question–document pair.
- Derived computable retrieval-error bounds using a multivariate Gaussian approximation of TF–IDF similarity margins.
- Implemented and evaluated the VALID Byzantine-robust decentralized learning algorithm (ISIT 2024) in PyTorch, preserving standard learning performance in adversary-free settings and achieving 30–50% higher accuracy than state-of-the-art robust defenses on MNIST and CIFAR-10.
- Designed and evaluated two schemes for decentralized sparse matrix multiplication under Byzantine attacks (TIFS 2025), eliminating the need for a trusted server through Common Tag verification and validating robustness and system trade-offs across 1,000 simulation rounds.

Futurewei Technologies — Basking Ridge, NJ
Machine Learning Research Intern — *May 2025 – Aug 2025*

- Developed a 6G energy–latency optimization framework (Networking Letters 2026) combining adaptive per-user QoS recommendations with UP-path selection and transmit-power control.
- Trained a three-layer PyTorch model on 20,000 3GPP-aligned samples and integrated its recommendations with Gurobi optimization, reducing simulated energy consumption by 18–25% versus fixed-latency baselines and 40–50% versus non-energy-aware systems while lowering latency by 30–40%.

Ozyegin University — Istanbul, Turkey
Research Assistant — *Jan 2018 – Jul 2021*

- Extended simulated underwater-sensor battery lifetime by 15–110%, depending on spectral-efficiency requirements, by optimizing time-switching, power-splitting, and hybrid SLIPT strategies.
- Enabled constrained energy optimization under underwater turbulence by deriving closed-form expressions for harvested energy, bit-error rate, and spectral efficiency and solving for optimal switching and splitting factors.
- Published the work in IEEE Transactions on Wireless Communications as corresponding author after leading the analysis, simulation code, figure generation, and manuscript development.

Selected Projects

Parameter-Efficient NLP Fine-Tuning
PyTorch, Hugging Face Transformers, PEFT, LoRA

- Achieved approximately 92% SST-2 validation accuracy by fine-tuning DistilBERT for sentiment classification with rank-8 LoRA while updating approximately 0.8% of model parameters.
- Published a reusable 2.96 MB adapter with tokenizer, configuration, and model card on the Hugging Face Hub using a reproducible three-epoch training and evaluation pipeline.

AI Job Scout Agent

Vibe Coding, Python, Gemini API

- Built an LLM-powered job-search agent that aggregates postings from multiple sources, applies embedding-based semantic ranking, and stores deduplicated results in SQLite.
- Extended with agentic deployment patterns using Google's Agent Development Kit (ADK) and Vertex AI (Google AI Agents Workshop, Jun 2026).

Education

New Jersey Institute of Technology — Newark, NJ
Ph.D., Electrical Engineering — *Jan 2022 – May 2026*
Research Focus: Reliable Machine Learning and Information Retrieval under Uncertainty
Advisor: *Prof. Joerg Kliewer*

Ozyegin University — Istanbul, Turkey
M.Sc., Electrical Engineering — *Feb 2018– Jul 2021*
Research Focus: Optimization, Resource Allocation, and Wireless Communications
Advisor: *Prof. Murat Uysal*

Selected Publications

Full list on **Google Scholar**.

1. **S. Ghasvarianjahromi**, J. Barr, Y. Yakimenka, J. Kliewer, "Context-Aware Search and Retrieval Under Token Erasure," under review, 2026.
2. **S. Ghasvarianjahromi**, A. Kiani, A. Xiang, et al., "AI/ML-Based QoS Recommendations for Energy Optimization in 6G Networks," *IEEE Networking Letters*, 2026.
3. **S. Ghasvarianjahromi**, Y. Yakimenka, J. Kliewer, "Decentralized Sparse Matrix Multiplication Under Byzantine Attacks," *IEEE Transactions on Information Forensics and Security*, 2025.
4. M. Bakshi, **S. Ghasvarianjahromi**, et al., "VALID: A Validated Algorithm for Learning in Decentralized Networks with Possible Adversarial Presence," *IEEE ISIT*, 2024.

Professional Service

IEEE Reviewer: Wireless Communications Letters; Communications Letters; Transactions on Communications; Transactions on Vehicular Technology; Transactions on Wireless Communications; Elsevier Physical Communication